



United International University

QUEST FOR EXCELLENCE

BiniOrbit

Connect | Invest | Grow

Database Management Systems Laboratory

Section: G

Submitted by:

Group: 1, IDTS

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1. Introduction

BiniOrbit — a platform built to connect startups and investors in Bangladesh. It is a comprehensive web-based platform designed to bridge the gap between entrepreneurs and investors in Bangladesh's growing startup ecosystem. The platform serves as a centralized hub where business owners can showcase their ventures and investors can discover promising investment opportunities.

The application facilitates meaningful connections through an intuitive interface that combines social networking features with investment-focused tools. By leveraging modern web technologies and a robust database system, BiniOrbit creates a secure environment for business discovery, communication, and investment facilitation.

Key stakeholders include:

Entrepreneurs: Seeking funding and visibility for their startups

Investors: Looking for credible early-stage investment opportunities

2. Motivation

We are motivated to build BiniOrbit because we see both the problems and the opportunities:

- **Digital Bangladesh:** With more than **100 million internet users**, the environment is perfect for an online platform like ours.
- **Investment Gap:** Only **2.6%** of startup funding in 2024 came from local investors — we want to change that.
- **Need for Scalability:** A separate login for Messenger ensures that startups going viral won't crash the whole system.
- **Young Population:** Bangladesh's youth are smart, energetic, and full of ideas. They deserve better support to grow.

- **Trust and Security:** In today's world, platforms must be safe and secure for everyone involved.

We believe BiniOrbit can help connect dreams with resources in a way that's simple, scalable, and secure.

3. Similar Projects

Existing Platforms Analysis

AngelList (International)

- **Focus:** Startup fundraising and recruiting
- **Features:** Company profiles, investor connections, job listings
- **Limitation:** Limited presence in Bangladesh market

Seedrs (Europe)

- **Focus:** Equity crowdfunding
- **Features:** Investment campaigns, due diligence tools
- **Limitation:** Geographic restrictions

Biniyog.io (Bangladesh)

- **Focus:** Shariah-compliant business financing and investment
- **Features:** Halal investment opportunities, Murabaha contracts, pre-vetted businesses, transparent repayment tracking
- **Limitation:** Focus on Islamic finance only, not equity-based investments like traditional startup funding

Biniyog Briddhi/SIE-B (Bangladesh)

- **Focus:** Scaling impact enterprises and social entrepreneurship
- **Features:** Capacity building for impact entrepreneurs, catalytic funding, investment readiness programs, gender lens integration
- **Limitation:** Focus on impact enterprises only, not general startup funding; program-based with specific application periods

Local Alternatives

- Limited dedicated platforms in Bangladesh
- Most funding connections happen through personal networks
- Existing business directories lack investment-focused features

4. Benchmark Analysis

Features	BiniOrbit	Biniyog.io	Biniyog Briddhi
User Registration (Investor/Business)	✓	✓	✗
Profile Listing	✓	✓	✓
Investment Request System	✓	✓	✗
Dedicated Messenger Login	✓	✗	✗
Real-Time Chat	✓	✗	✗
Scalable for Viral Users	✓	✗	✗
Admin Dashboard	✓	✗	✗
Verification System	✓	✓	✓
Connection Request	✓	✗	✗
Bangladeshi Regulatory Compliance	✓	✓	✓
Simple easy to understand UI	✓	✗	✗

5. Complete Feature List

1. User Management System

Task 1.1: User Registration (Business Owners)

- Business profile creation with verification documents
- TIN, VAT, and business registration validation
- Document upload (bank certificates, licenses)

Task 1.2: User Registration (Investors)

- Personal profile creation with investment preferences
- Investor type classification (Strategic, Angel, etc.)
- Investment history and portfolio tracking

Task 1.3: Authentication System

- Secure login/logout functionality
- Session management with 30-day persistence
- Password encryption and validation

2. Social Networking Features

Task 2.1: Newsfeed System

- Social media-style post creation and display
- Real-time post updates with AJAX
- Post interaction (like, comment, share)

Task 2.2: Professional Connections

- LinkedIn-style connection requests

- Connection status management (pending, accepted, removed)
- Network visualization and management

Task 2.3: User Profiles

- Comprehensive profile pages for businesses and investors
- Profile visiting and interaction tracking
- Public/private profile settings

3. Investment Management System

Task 3.1: Investment Request Creation

- Investment proposal submission with amount and equity
- Automatic business valuation calculation
- Investment type classification

Task 3.2: Investment Tracking

- Status management (pending, accepted, declined)
- Investment history and portfolio view
- Performance metrics and analytics

Task 3.3: Business Investment Dashboard

- Incoming investment request management
- Request approval/rejection workflow
- Investment portfolio overview

4. Communication System

Task 4.1: Real-time Messaging

- BiniOrbit Messenger integration

- Direct messaging between users
- Message history and persistence

Task 4.2: Notification System

- Real-time notifications for activities
- Email and in-app notification delivery
- Notification preferences management

5. Content Management

Task 5.1: Post Management

- Create, edit, and delete posts
- Image upload and management

Task 5.2: Comment System

- Comment creation and moderation
- Comment reporting and management
- Admin comment management tools

6. Admin Panel

Task 6.1: User Management

- View and manage all platform users
- User account status management
- User activity monitoring

Task 6.2: Content Moderation

- Comment reporting management
- Content approval workflows

- Platform policy enforcement

Task 6.3: Analytics Dashboard

- Platform usage statistics
- Investment transaction reports
- User engagement metrics

7. Search and Discovery

Task 7.1: Global Search

- Search across businesses and investors
- Advanced search filters
- Search result ranking

Task 7.2: Business Discovery

- Business category browsing
- Investment opportunity discovery
- Featured business highlighting

6. Database Design Approach

Design Philosophy:

- **Normalization:** Third Normal Form (3NF) compliance
- **Scalability:** Modular table design for future expansion
- **Performance:** Optimized indexing and query structure
- **Data Integrity:** Foreign key constraints and validation
- **Security:** Encrypted sensitive data storage

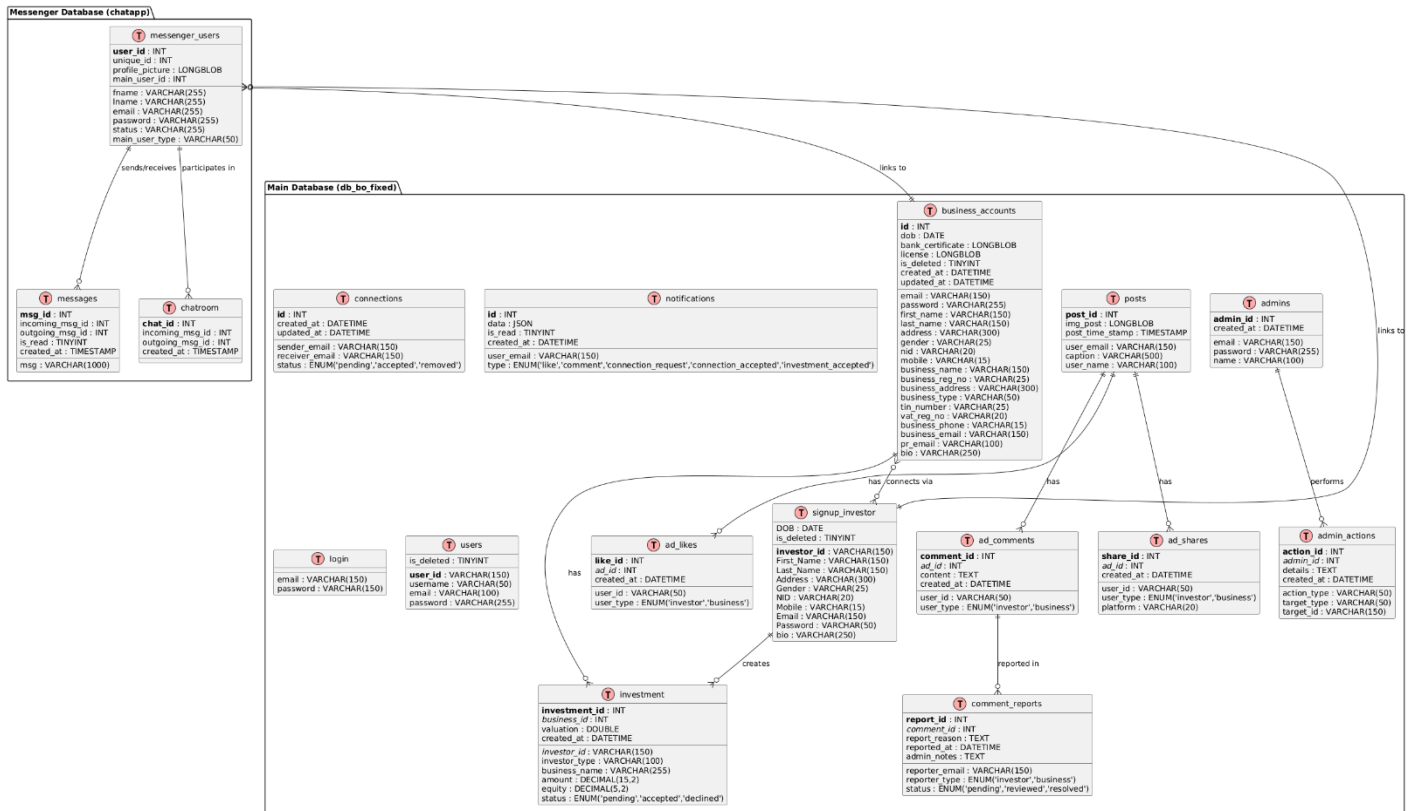
Database Architecture:

- **Primary Database:** db_bo_fixed (Main application data)
- **Messenger Database:** chatapp (Communication system)
- **Backup System:** Automated database backups
- **Migration Support:** Version-controlled schema changes

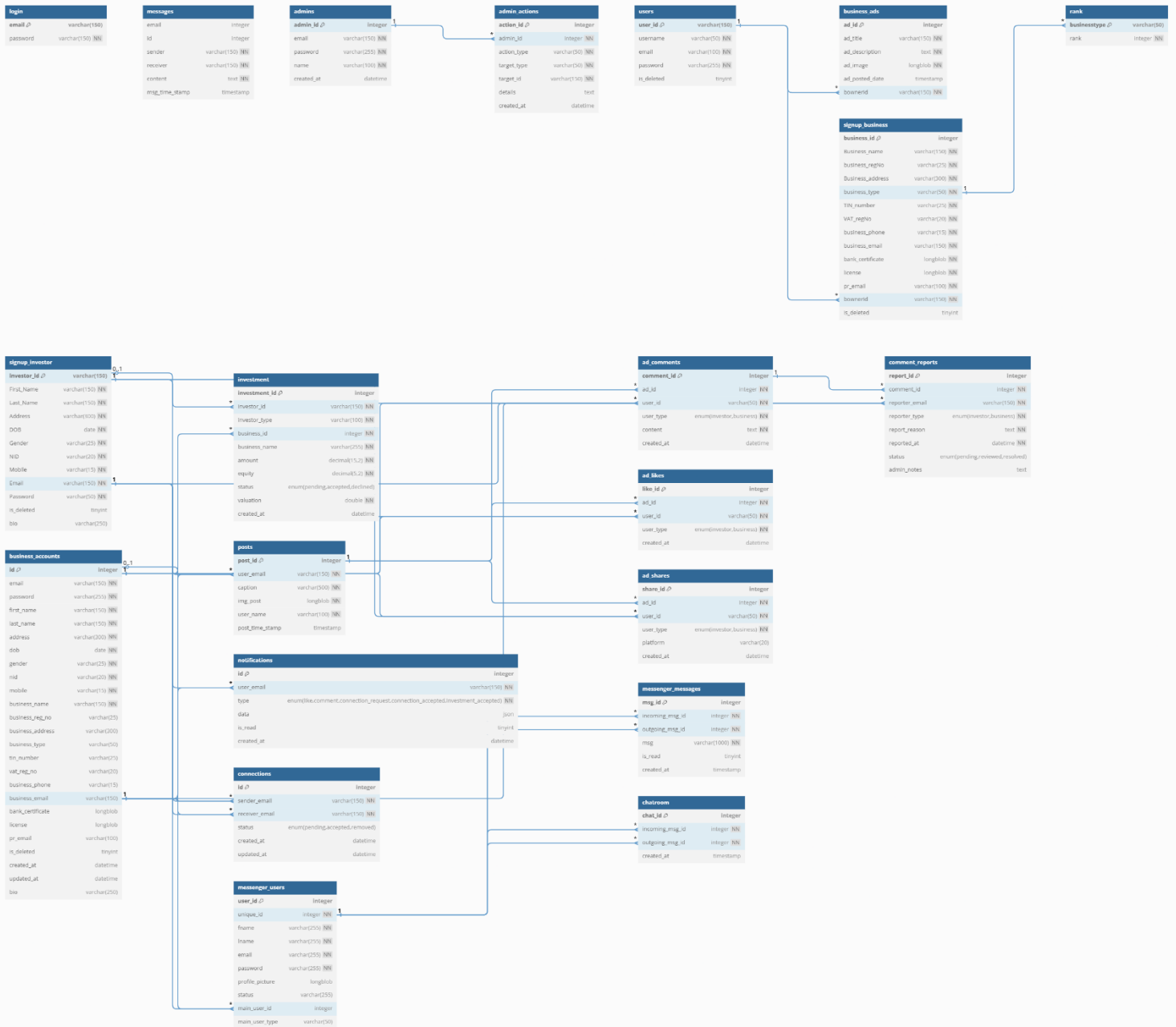
Design Principles:

1. **Separation of Concerns:** Different databases for different functionalities
2. **Data Consistency:** ACID compliance for transactions
3. **Performance Optimization:** Strategic indexing and query optimization
4. **Security First:** Input validation and SQL injection prevention
5. **Maintainability:** Clear naming conventions and documentation

7. Entity Relationship Diagram



8. Schema Diagram



9. Queries

1. User Authentication Queries

-- Investor Login

```
SELECT * FROM signup_investor
WHERE Email = ? AND Password = ? AND is_deleted = 0;
```

-- Business Login

```
SELECT * FROM business_accounts
WHERE business_email = ? AND password = ? AND is_deleted = 0;
```

-- Admin Login

```
SELECT * FROM admins
WHERE email = ? AND password = ?;
```

2. Investment Management Queries

-- Create Investment Request

```
INSERT INTO investment (investor_id, investor_type, business_id, business_name,
                        amount, equity, valuation, status, created_at)
VALUES (?, ?, ?, ?, ?, ?, ?, 'pending', NOW());
```

-- Get Investment Requests for Business

```
SELECT i.*, si.First_Name, si.Last_Name
FROM investment i
JOIN signup_investor si ON i.investor_id = si.investor_id
WHERE i.business_id = ?
ORDER BY i.created_at DESC;
```

-- Get Investor's Investment History

```
SELECT i.*, ba.business_name
FROM investment i
LEFT JOIN business_accounts ba ON i.business_id = ba.id
WHERE i.investor_id = ?
ORDER BY i.created_at DESC;
```

-- Update Investment Status

```
UPDATE investment
SET status = ?
WHERE investment_id = ?;
```

3. Social Networking Queries

-- Create Connection Request

```
INSERT INTO connections (sender_email, receiver_email, status, created_at)
VALUES (?, ?, 'pending', NOW());
```

-- Get User Connections

```
SELECT * FROM connections
WHERE (sender_email = ? OR receiver_email = ?)
AND status = 'accepted';
```

-- Get Pending Connection Requests

```
SELECT * FROM connections
WHERE receiver_email = ? AND status = 'pending';
```

-- Accept Connection Request

```
UPDATE connections
SET status = 'accepted', updated_at = NOW()
WHERE id = ?;
```

4. Post and Comment Queries

-- Get Newsfeed Posts with Engagement Counts

```
SELECT p.*,
       (SELECT COUNT(*) FROM ad_likes WHERE ad_likes.ad_id = p.post_id) as
like_count,
       (SELECT COUNT(*) FROM ad_comments WHERE ad_comments.ad_id =
p.post_id) as comment_count,
       (SELECT COUNT(*) FROM ad_shares WHERE ad_shares.ad_id = p.post_id) as
share_count,
       (SELECT COUNT(*) FROM ad_likes WHERE ad_likes.ad_id = p.post_id
       AND ad_likes.user_id = ? AND ad_likes.user_type = ?) as user_liked
FROM posts p
ORDER BY p.post_time_stamp DESC;
```

-- Add Comment

```
INSERT INTO ad_comments (ad_id, user_id, user_type, content, created_at)
VALUES (?, ?, ?, ?, NOW());
```

-- Get Comments with User Information

```
SELECT c.*,
       CASE
         WHEN c.user_type = 'investor' THEN
           (SELECT CONCAT(First_Name, ' ', Last_Name) FROM signup_investor
           WHERE investor_id = c.user_id)
         WHEN c.user_type = 'business' THEN
```

```
        (SELECT business_name FROM business_accounts WHERE id = c.user_id)
    END as user_name
FROM ad_comments c
WHERE c.ad_id = ?
ORDER BY c.created_at DESC;
```

5. Messaging Queries

-- Get Chat Messages

```
SELECT m.*, u.fname, u.lname
FROM messages m
LEFT JOIN users u ON u.unique_id = m.outgoing_msg_id
WHERE (m.incoming_msg_id = ? AND m.outgoing_msg_id = ?)
    OR (m.incoming_msg_id = ? AND m.outgoing_msg_id = ?)
ORDER BY m.created_at ASC;
```

-- Insert New Message

```
INSERT INTO messages (incoming_msg_id, outgoing_msg_id, msg, created_at)
VALUES (?, ?, ?, NOW());
```

-- Get Unread Message Count

```
SELECT outgoing_msg_id, COUNT(*) as unread_count
FROM messages
WHERE incoming_msg_id = ? AND is_read = 0
GROUP BY outgoing_msg_id;
```

6. Notification Queries

-- Create Notification

```
INSERT INTO notifications (user_email, type, data, created_at)
VALUES (?, ?, ?, NOW());
```

-- Get User Notifications

```
SELECT id, type, data, is_read, created_at
FROM notifications
WHERE user_email = ?
ORDER BY created_at DESC;
```

-- Mark Notification as Read

```
UPDATE notifications
SET is_read = 1
WHERE id = ?;
```

7. Search and Discovery Queries

-- Search Businesses

```
SELECT * FROM business_accounts
WHERE business_name LIKE ? OR business_type LIKE ?
AND is_deleted = 0;
```

-- Search Investors

```
SELECT * FROM signup_investor
WHERE First_Name LIKE ? OR Last_Name LIKE ?
AND is_deleted = 0;
```

-- Get Business by Type

```
SELECT * FROM business_accounts
WHERE business_type = ? AND is_deleted = 0;
```

8. Admin Management Queries

-- Get All Users


```
SELECT 'investor' as type, investor_id as id, First_Name, Last_Name, Email
FROM signup_investor WHERE is_deleted = 0
UNION ALL
SELECT 'business' as type, id, first_name, last_name, business_email as Email
FROM business_accounts WHERE is_deleted = 0;
```

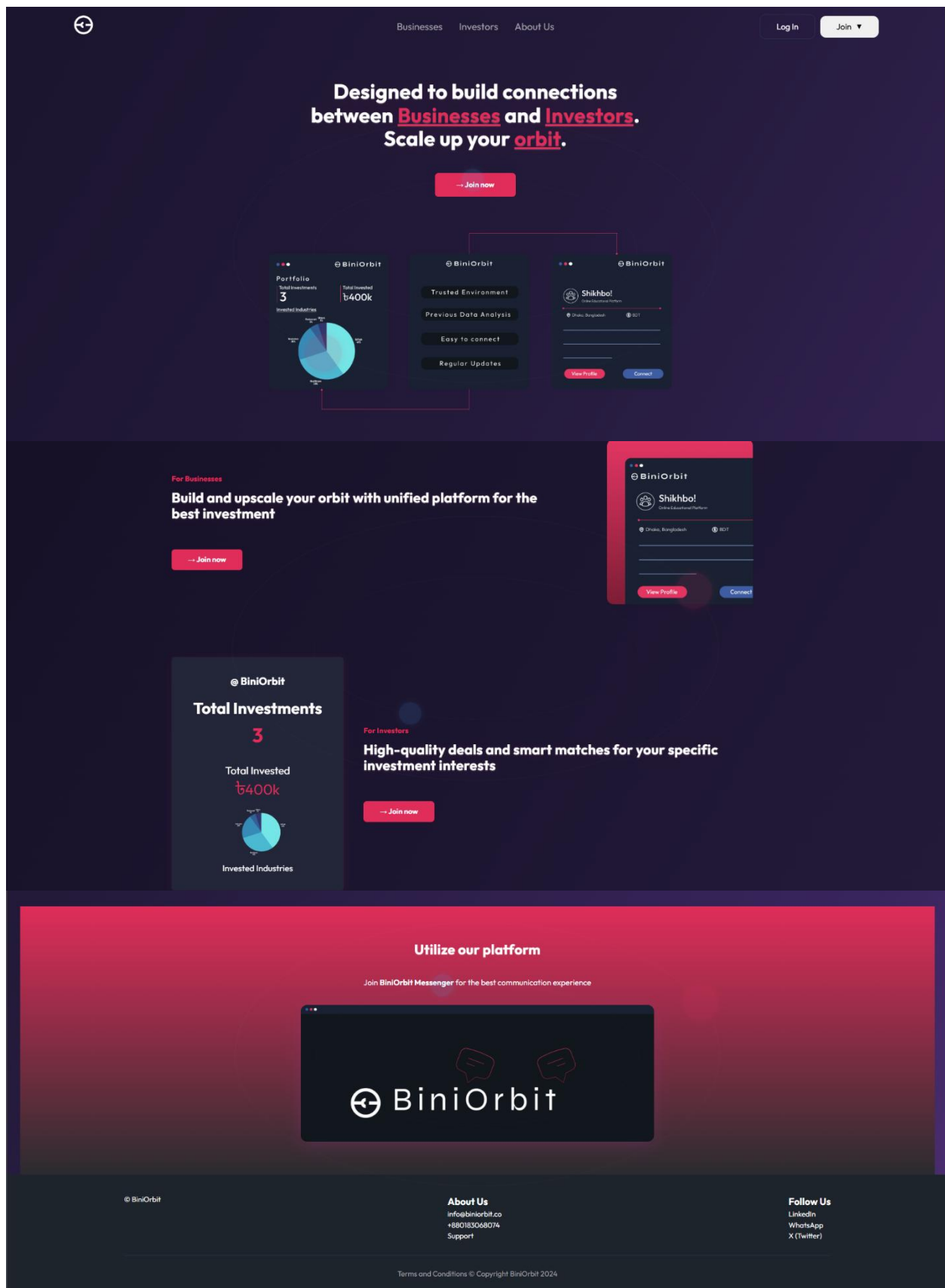
-- Get Reported Comments

```
SELECT cr.*, c.content, c.user_id, c.user_type
FROM comment_reports cr
JOIN ad_comments c ON cr.comment_id = c.comment_id
WHERE cr.status = 'pending'
ORDER BY cr.reported_at DESC;
```

-- Log Admin Action

```
INSERT INTO admin_actions (admin_id, action_type, target_type, target_id, details,
created_at)
VALUES (?, ?, ?, ?, ?, NOW());
```

10. Key User Interface



BepariHouse

Newsfeed

Messenger

Notifications2

Profile

Connections

Investment Requests

Logout

Support

Theme

Search users...

What's on your mind?

Add Image

Post

BepariHouse

Jul 2, 2025 8:46 AM

Invest with us

Let Your Finances Soar



Invest with us today

BepariHouse

Jul 2, 2025 8:46 AM

Invest with us

Let Your Finances Soar



Invest with us today

Like 2

Comment 1

Share 0

Write a comment...

Atique Mahmud

Interested

LikeReplyReport

Just now

Notifications

Atique Mahmud commented on your post.

1 minute ago

Atique Mahmud liked your post.

2 minutes ago

Someone accepted your connection request.

2 days ago

BepariHouse

beparih@gmail.com

1

1

1

Posts

Connections

Investments

Edit Profile

View Investments

Share your thoughts...

Add Media

Post

BepariHouse

July 2, 2025

Invest with us

Profile Information

Email

beparih@gmail.com

Member Since

July 2025

Location

Not specified

Connections

Connection Requests

No pending connection requests

Connected

Atique Mahmud

atq@gmail.com

Investor

View Profile

Disconnect

Make an Investment

Connect with businesses and grow your portfolio

[← Back to Newsfeed](#)

Investment Type

Please select your investment type

Strategic

Advisory

Business Selection

Investing in: Select a business

Business Name

Search or type a business name...

Investment Amount

Quick Select Amount

₹10,000

₹20,000

₹50,000

₹100,000

₹500,000

₹1,000,000

Custom Amount

₹ Enter amount in BDT

Equity Share

10%

Adjust the slider to set your desired equity percentage

Estimated Business Valuation:

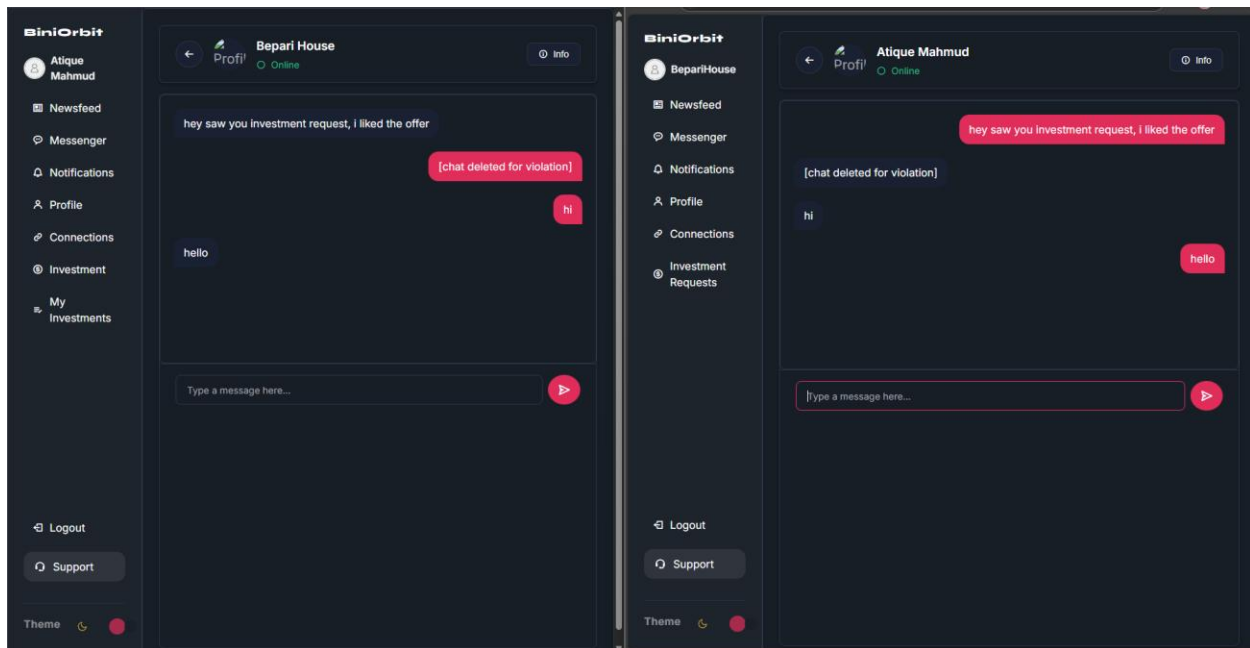
₹0

Calculated as: Investment Amount / (Equity % / 100)

Important Disclaimer

Investments are subject to market risks. Please read all related documents carefully before investing. Past performance is not indicative of future results. Always consult with a financial advisor before making investment decisions.

[Submit Investment](#)



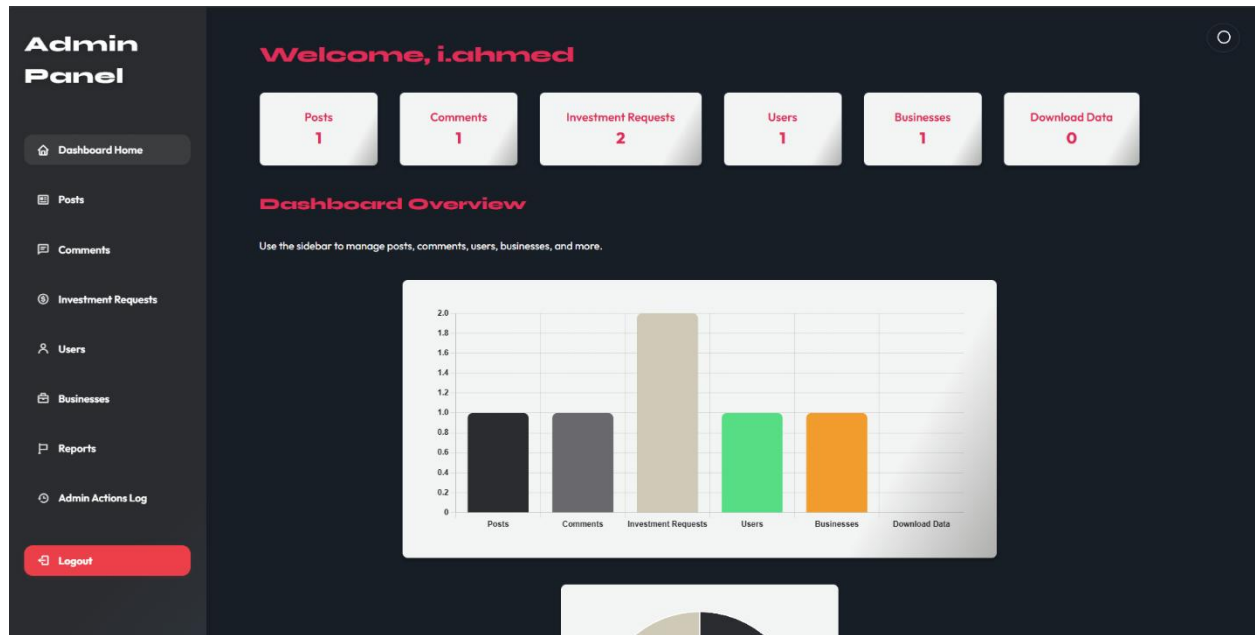
💰 My Investments

Business	Type	Amount	Equity	Status	Action
BepariHouse	Strategic	₹20,000.00	20%	Pending	-
BepariHouse	Strategic	₹50,000.00	11%	Accepted	-

+ New Investment

💰 Investment Requests

Investor	Email	Type	Amount	Equity	Valuation	Status	Action
Atique Mahmud	atq@gmail.com	Strategic	₹20,000.00	20%	₹100,000.00	Pending	 
Atique Mahmud	atq@gmail.com	Strategic	₹50,000.00	11%	₹454,545.45	Accepted	



11. Limitations

Technical Limitations:

1. **Single Database Server:** Currently uses local XAMPP setup
2. **File Upload Size:** Limited by PHP configuration
3. **Real-time Updates:** Basic AJAX implementation without WebSockets
4. **Mobile App:** Web-only platform, no native mobile application
5. **Payment Integration:** No direct payment processing system

Functional Limitations:

1. **Geographic Scope:** Limited to local market initially
2. **Verification Process:** Manual document verification
3. **Advanced Analytics:** Basic reporting without advanced insights

4. **API Integration:** Limited third-party integrations
5. **Multi-language Support:** English-only interface

Security Limitations:

1. **Two-Factor Authentication:** Not implemented
2. **Advanced Encryption:** Basic password hashing
3. **API Security:** No comprehensive API security measures
4. **Audit Trail:** Limited system-wide audit logging

12. Future Work

Short-term Enhancements (3-6 months):

1. **Enhanced Security**
 - Implement two-factor authentication
 - Advanced encryption for sensitive data
 - Comprehensive security audit
1. **Mobile Application**
 - Native iOS and Android apps
 - Push notifications
 - Offline functionality
1. **Payment Integration**
 - Stripe/PayPal integration
 - Escrow services for investments
 - Transaction fee management

Medium-term Improvements (6-12 months):

1. **Advanced Analytics**
 - Investment performance tracking

- Business growth metrics

1. **AI Integration**

- Smart investment matching
- Automated content moderation
- Chatbot support

1. **API Development**

- RESTful API for third-party integrations
- Webhook support
- Developer documentation

Long-term Vision (1-2 years):

1. **Global Expansion**

- Multi-language support
- International payment systems
- Regional compliance features

1. **Advanced Features**

- Blockchain integration for transparency
- Smart contracts for investments
- Decentralized identity verification

1. **Ecosystem Development**

- Partner integrations
- Third-party service marketplace
- Community features

Technical Roadmap:

1. **Scalability Improvements**

- Cloud deployment
- Load balancing
- Database optimization

13. Conclusion

BiniOrbit represents a comprehensive solution for business-investor networking, successfully combining social media features with investment management capabilities. The platform demonstrates advanced web development practices, robust database design, and user-centric interface development.

Key Achievements:

1. **Complete Platform Integration:** Seamless integration of multiple functionalities
2. **Advanced Investment System:** Automated valuation and comprehensive tracking
3. **Real-time Communication:** Instant messaging and notification systems
4. **Professional Networking:** LinkedIn-style connection management
5. **Content Management:** Social media-style posting and interaction

Technical Excellence:

- **Modern Development Stack:** PHP 8.2, MySQL, HTML5/CSS3/JavaScript
- **Security Implementation:** Session management, input validation, SQL injection prevention
- **Database Design:** Normalized schema with proper relationships and constraints
- **User Experience:** Responsive design with modern UI/UX principles
- **Performance Optimization:** Efficient queries and proper indexing